

Class 18

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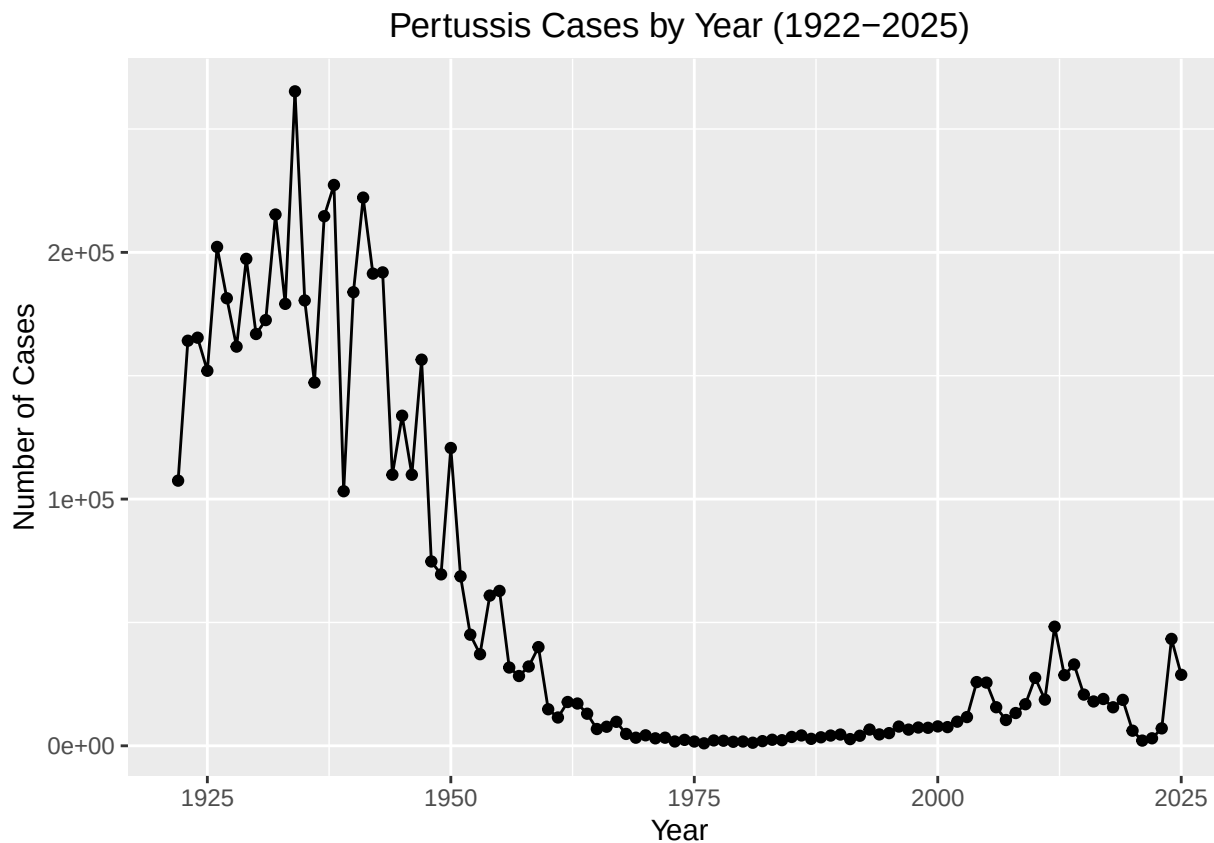
Background

```
cdc <- data.frame(
  year = c(1922L,1923L,1924L,1925L,
           1926L,1927L,1928L,1929L,1930L,1931L,
           1932L,1933L,1934L,1935L,1936L,
           1937L,1938L,1939L,1940L,1941L,1942L,
           1943L,1944L,1945L,1946L,1947L,
           1948L,1949L,1950L,1951L,1952L,
           1953L,1954L,1955L,1956L,1957L,1958L,
           1959L,1960L,1961L,1962L,1963L,
           1964L,1965L,1966L,1967L,1968L,1969L,
           1970L,1971L,1972L,1973L,1974L,
           1975L,1976L,1977L,1978L,1979L,1980L,
           1981L,1982L,1983L,1984L,1985L,
           1986L,1987L,1988L,1989L,1990L,
           1991L,1992L,1993L,1994L,1995L,1996L,
           1997L,1998L,1999L,2000L,2001L,
           2002L,2003L,2004L,2005L,2006L,2007L,
           2008L,2009L,2010L,2011L,2012L,
           2013L,2014L,2015L,2016L,2017L,2018L,
           2019L,2020L,2021L,2022L,2023L,2024L,2025L),
  No..Reported.Pertussis.Cases = c(107473,164191,165418,152003,
                                   202210,181411,161799,197371,
                                   166914,172559,215343,179135,265269,
                                   180518,147237,214652,227319,103188,
                                   183866,222202,191383,191890,109873,
                                   133792,109860,156517,74715,69479,
                                   120718,68687,45030,37129,60886,
                                   62786,31732,28295,32148,40005,
                                   14809,11468,17749,17135,13005,6799,
                                   7717,9718,4810,3285,4249,3036,
                                   3287,1759,2402,1738,1010,2177,2063,
                                   1623,1730,1248,1895,2463,2276,
                                   3589,4195,2823,3450,4157,4570,
                                   2719,4083,6586,4617,5137,7796,6564,
                                   7405,7298,7867,7580,9771,11647,
                                   25827,25616,15632,10454,13278,
                                   16858,27550,18719,48277,28639,32971,
                                   20762,17972,18975,15609,18617,
                                   6124,2116,3044,7063,43321,28783)
)
```

Q1. With the help of the R “addin” package `datapasta` assign the CDC pertussis case number data to a data frame called `cdc` and use `ggplot` to make a plot of cases numbers over time.

```
cdc$cases <- cdc$No..Reported.Pertussis.Cases
library(ggplot2)

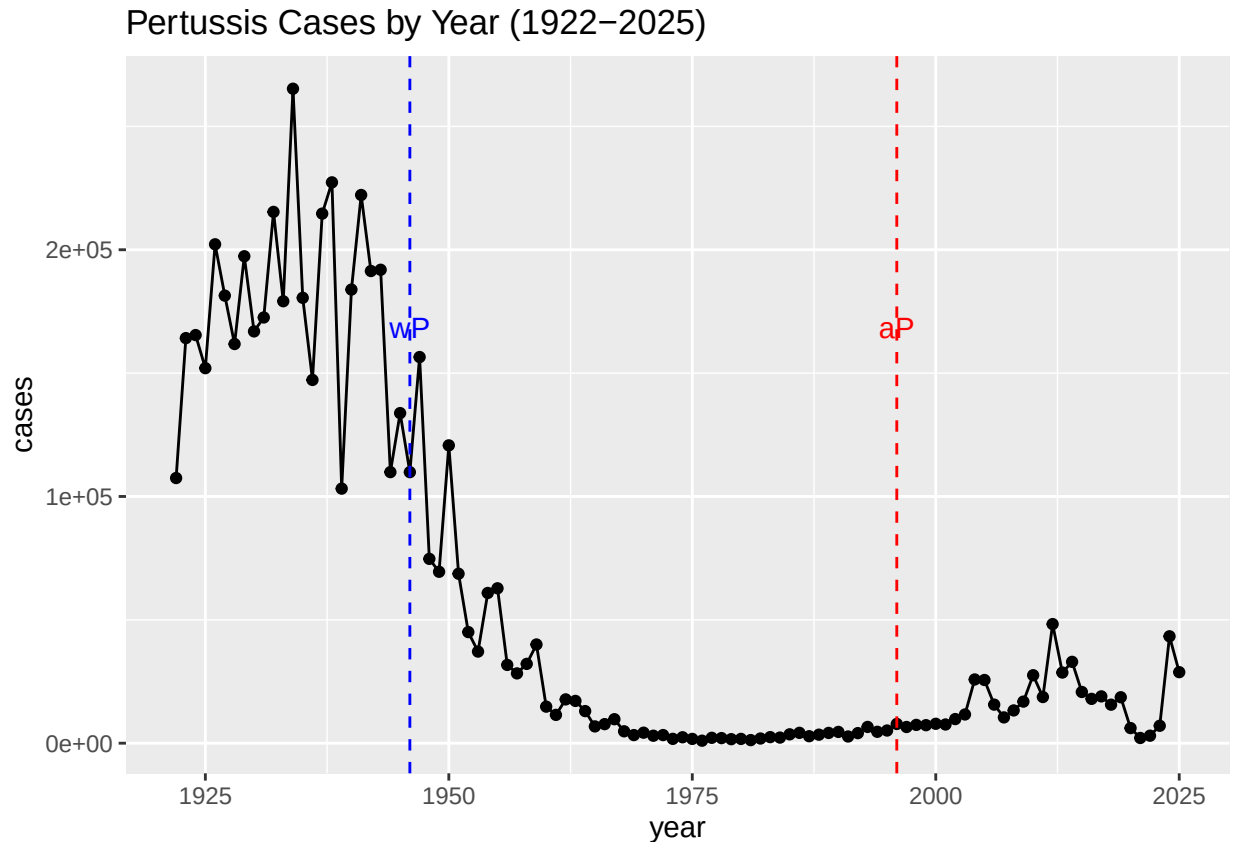
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  labs(title = "Pertussis Cases by Year (1922-2025)",
       x = "Year",
       y = "Number of Cases") +
  theme(plot.title = element_text(hjust = 0.5))
)
```



Q2. Using the `ggplot` `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc) +
  aes(x = year, y = cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, col = "blue", linetype = "dashed") +
  geom_vline(xintercept = 1996, col = "red", linetype = "dashed") +
  annotate("text", x = 1946, y = 160000, label = "wP", col = "blue", vjust = -0.5) +
```

```
annotate("text", x = 1996, y = 160000, label = "aP", col = "red", vjust = -0.5) +
labs(title = "Pertussis Cases by Year (1922-2025)")
```



After the wP vaccine was introduced around 1946, pertussis cases dropped dramatically from very high levels to much lower levels. After the switch to the aP vaccine around 1996, cases started to slowly increase again, although they stayed much lower than before vaccination. This suggests that the wP vaccine greatly reduced pertussis cases, while the aP vaccine may not provide protection that lasts as long.

Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend? After the wP, or whole-cell pertussis vaccine, was introduced in the mid-1940s, the number of pertussis cases dropped a lot, from over 200,000 cases to only hundreds of cases.

After the switch to the newer aP, or acellular pertussis vaccine, around 1996, pertussis cases slowly started to rise again. One possible reason is that the protection from the aP vaccine does not last as long as the older wP vaccine.

In 2020, COVID-19 lockdowns and social distancing also caused pertussis cases to drop again. This shows that measures used to stop COVID-19 transmission also helped reduce the spread of pertussis.

Exploring CMI-PB data

```
library(jsonlite)

subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)

head(subject,3)
```

```
##   subject_id infancy_vac biological_sex      ethnicity race
## 1          1          wP      Female Not Hispanic or Latino White
## 2          2          wP      Female Not Hispanic or Latino White
## 3          3          wP      Female      Unknown White
##   year_of_birth date_of_boost      dataset
## 1   1986-01-01   2016-09-12 2020_dataset
## 2   1968-01-01   2019-01-28 2020_dataset
## 3   1983-01-01   2016-10-10 2020_dataset
```

Q.4 How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```
##
## aP wP
## 87 85
```

- aP (Acellular Pertussis): 87 subjects
- wP (Whole-cell Pertussis): 85 subjects

Q5. How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

```
##
## Female    Male
##    112     60
```

Female: 112 Male: 60

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$race, subject$biological_sex)
```

```
##
##                                     Female Male
## American Indian/Alaska Native          0    1
## Asian                               32   12
## Black or African American             2    3
## More Than One Race                   15    4
## Native Hawaiian or Other Pacific Islander  1    1
## Unknown or Not Reported              14    7
## White                               48   32
```

The breakdown is not particularly representative of the US population - this is a serious caveat for this study. However, it is still the largest sample of it's type ever assembled.

Side-Note: Working with dates

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
library(lubridate)
```

```
##
## Attaching package: 'lubridate'
## The following objects are masked from 'package:base':
##
```

```
##      date, intersect, setdiff, union
library(dplyr)

##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##      filter, lag
## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union
```

```
subject$age <- today() - ymd(subject$year_of_birth)
```

```
ap <- subject %>% filter(infancy_vac == "aP")
round( summary( time_length( ap$age, "years" ) ) )
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      23      27      28      28      29      35
```

```
wp <- subject %>% filter(infancy_vac == "wP")
round( summary( time_length( wp$age, "years" ) ) )
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      23      33      35      37      40      58
```

The average age of the aP individuals is 28 years, while the average age of the wP individuals is 37 years. The wP group is older on average than the aP group. These two groups are significantly different, with an extremely small p-value, meaning the age difference is unlikely to be due to chance.

Q8. Determine the age of all individuals at time of boost?

```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

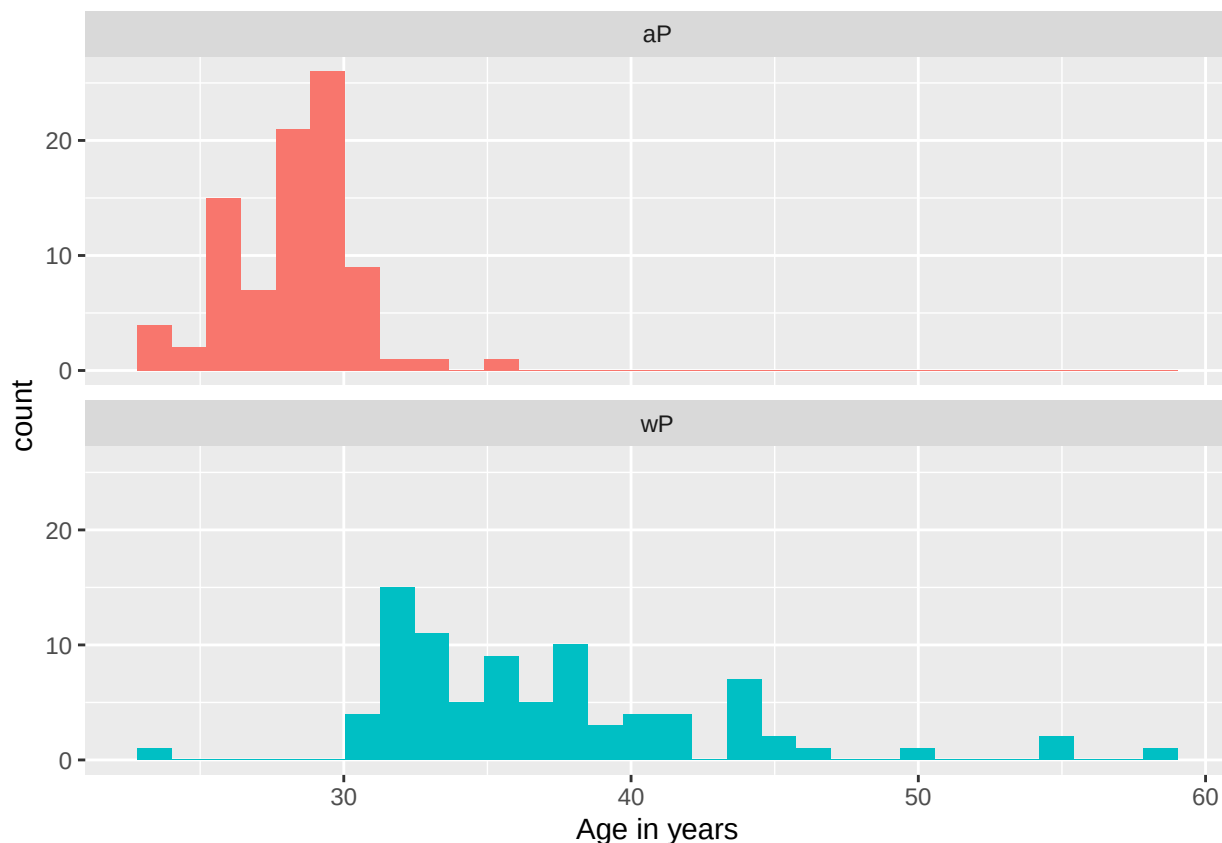
```
## [1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

The ages of the individuals at the time of boost are approximately 30.70, 51.07, 33.77, 28.66, 25.66, and 28.77 years old.

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

```
ggplot(subject) +
  aes(time_length(age, "year"),
      fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
# Or use wilcox.test()
x <- t.test(time_length( wp$age, "years" ),
            time_length( ap$age, "years" ))

x$p.value
```

```
## [1] 2.372101e-23
```

The age distributions for the aP and wP vaccine groups are clearly different. The wP group is generally older, while the aP group is mostly younger and clustered around the mid-20s. This difference is statistically significant, with an extremely small p-value, meaning it is very unlikely that the difference happened by chance.

Joining multiple tables

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
library(dplyr)
specimen <- read_json("https://www.cmi-pb.org/api/v5_1/specimen",
                      simplifyVector = TRUE)
ab_titer <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer",
                      simplifyVector = TRUE)

meta <- inner_join(specimen, subject)

## Joining with `by = join_by(subject_id)`
```

```
dim(meta)
```

```
## [1] 1503 14
```

```
head(meta)
```

```
## specimen_id subject_id actual_day_relative_to_boost
## 1          1          1                          -3
## 2          2          1                           1
## 3          3          1                           3
## 4          4          1                           7
## 5          5          1                          11
## 6          6          1                          32
## planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
## 1                             0          Blood    1          wP          Female
## 2                             1          Blood    2          wP          Female
## 3                             3          Blood    3          wP          Female
## 4                             7          Blood    4          wP          Female
## 5                            14          Blood    5          wP          Female
## 6                            30          Blood    6          wP          Female
## ethnicity race year_of_birth date_of_boost dataset
## 1 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 2 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 3 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 4 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 5 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 6 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## age
## 1 14761 days
## 2 14761 days
## 3 14761 days
## 4 14761 days
## 5 14761 days
## 6 14761 days
```

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
abdata <- inner_join(ab_titer, meta)
```

```
## Joining with `by` = join_by(specimen_id)`
```

```
dim(abdata)
```

```
## [1] 61956 21
```

The titer and metadata were successfully joined by specimen_id, creating a new combined dataset with 61956 rows and 21 columns for further analysis.

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(abdata$isotype)
```

```
##
## IgE IgG IgG1 IgG2 IgG3 IgG4
## 6698 7265 11993 12000 12000 12000
```

The dataset contains 6 antibody isotypes: IgE = 6,698, IgG = 7,265, IgG1 = 11,993, and IgG2, IgG3, and IgG4 each have 12,000 specimens.

Q12. What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

```
table(abdata$dataset)
```

```
##
## 2020_dataset 2021_dataset 2022_dataset 2023_dataset
##          31520          8085          7301          15050
```

The different dataset values are 2020_dataset, 2021_dataset, 2022_dataset, and 2023_dataset. The most recent dataset, 2023_dataset, has 15,050 rows, which is much larger than the 2021 and 2022 datasets but still smaller than the 2020 dataset.

Examine IgG Ab titer levels

IgG is crucial for long-term immunity and responding to bacterial & viral infections

```
IgG <- abdata |>
```

```
  filter(isotype == "IgG")
```

```
head(IgG)
```

```
## specimen_id isotype is_antigen_specific antigen MFI MFI_normalised
## 1          1      IgG                TRUE      PT  68.56614      3.736992
## 2          1      IgG                TRUE      PRN 332.12718      2.602350
## 3          1      IgG                TRUE      FHA 1887.12263     34.050956
## 4         19      IgG                TRUE      PT  20.11607      1.096366
## 5         19      IgG                TRUE      PRN 976.67419      7.652635
## 6         19      IgG                TRUE      FHA  60.76626      1.096457
## unit lower_limit_of_detection subject_id actual_day_relative_to_boost
## 1 IU/ML                0.530000          1                -3
## 2 IU/ML                6.205949          1                -3
## 3 IU/ML                4.679535          1                -3
## 4 IU/ML                0.530000          3                -3
## 5 IU/ML                6.205949          3                -3
## 6 IU/ML                4.679535          3                -3
## planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
## 1                0          Blood          1          wP          Female
## 2                0          Blood          1          wP          Female
## 3                0          Blood          1          wP          Female
## 4                0          Blood          1          wP          Female
## 5                0          Blood          1          wP          Female
## 6                0          Blood          1          wP          Female
## ethnicity race year_of_birth date_of_boost dataset
## 1 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 2 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 3 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 4          Unknown White 1983-01-01 2016-10-10 2020_dataset
## 5          Unknown White 1983-01-01 2016-10-10 2020_dataset
## 6          Unknown White 1983-01-01 2016-10-10 2020_dataset
## age
## 1 14761 days
## 2 14761 days
## 3 14761 days
## 4 15857 days
## 5 15857 days
```

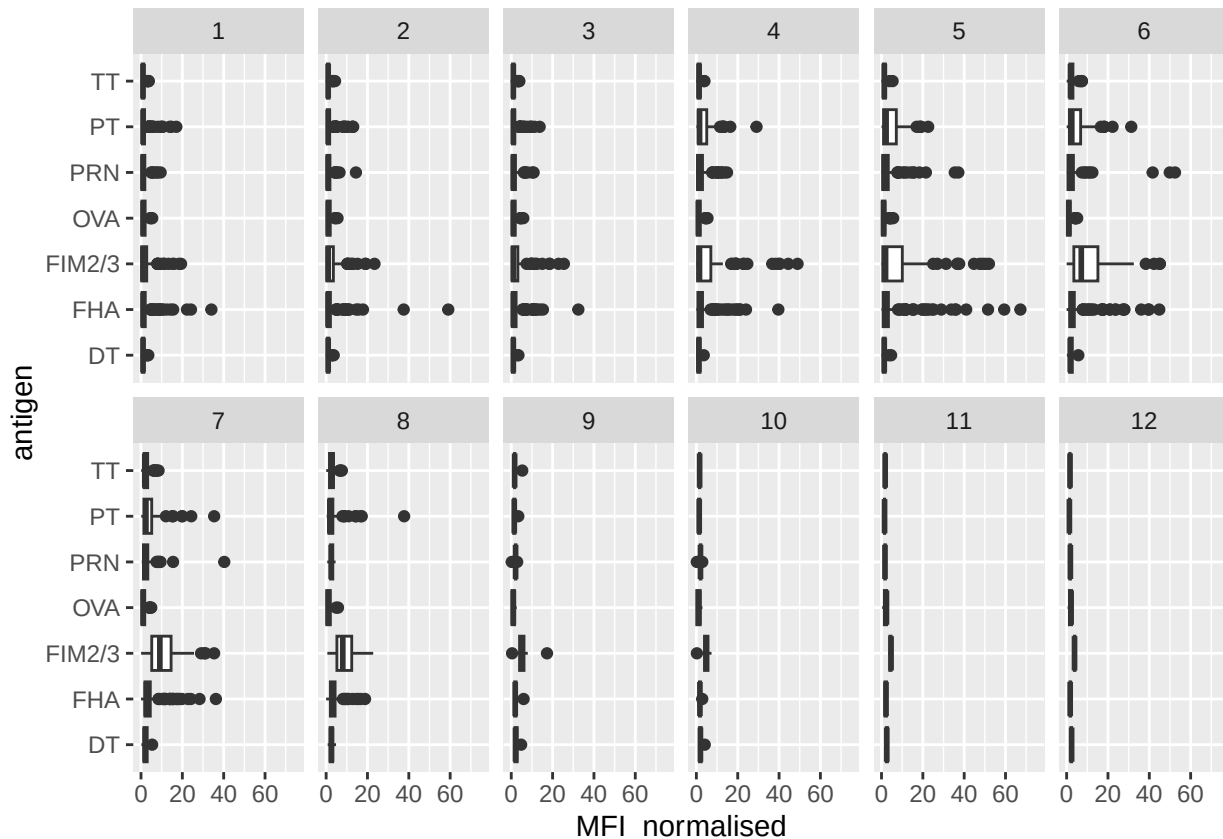


```
## 6 15857 days
```

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(IgG) +  
  aes(MFI_normalised, antigen) +  
  geom_boxplot() + xlim(0,75) +  
  facet_wrap(vars(visit), nrow=2)
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



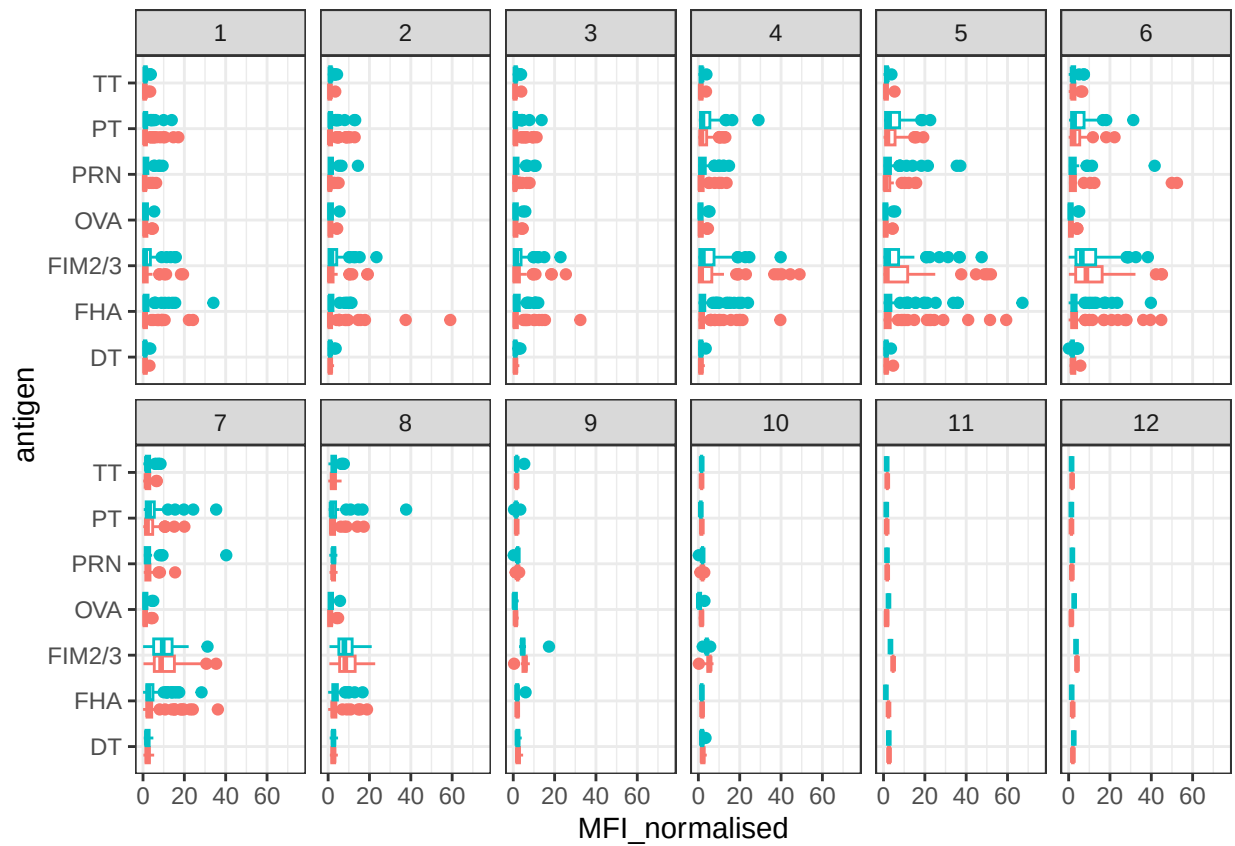
Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

The antigens PT, PRN, FHA, and FIM2/3 show clear differences in IgG antibody levels over time. This is likely because these are pertussis antigens that are included in the acellular pertussis vaccine, so the immune system specifically responds to them after vaccination. Other antigens, such as OVA, do not show the same pattern because they are not part of the pertussis vaccine and are used more like controls.

```
ggplot(IgG) +  
  aes(MFI_normalised, antigen, col=infancy_vac ) +  
  geom_boxplot(show.legend = FALSE) +  
  facet_wrap(vars(visit), nrow=2) +  
  xlim(0,75) +  
  theme_bw()
```

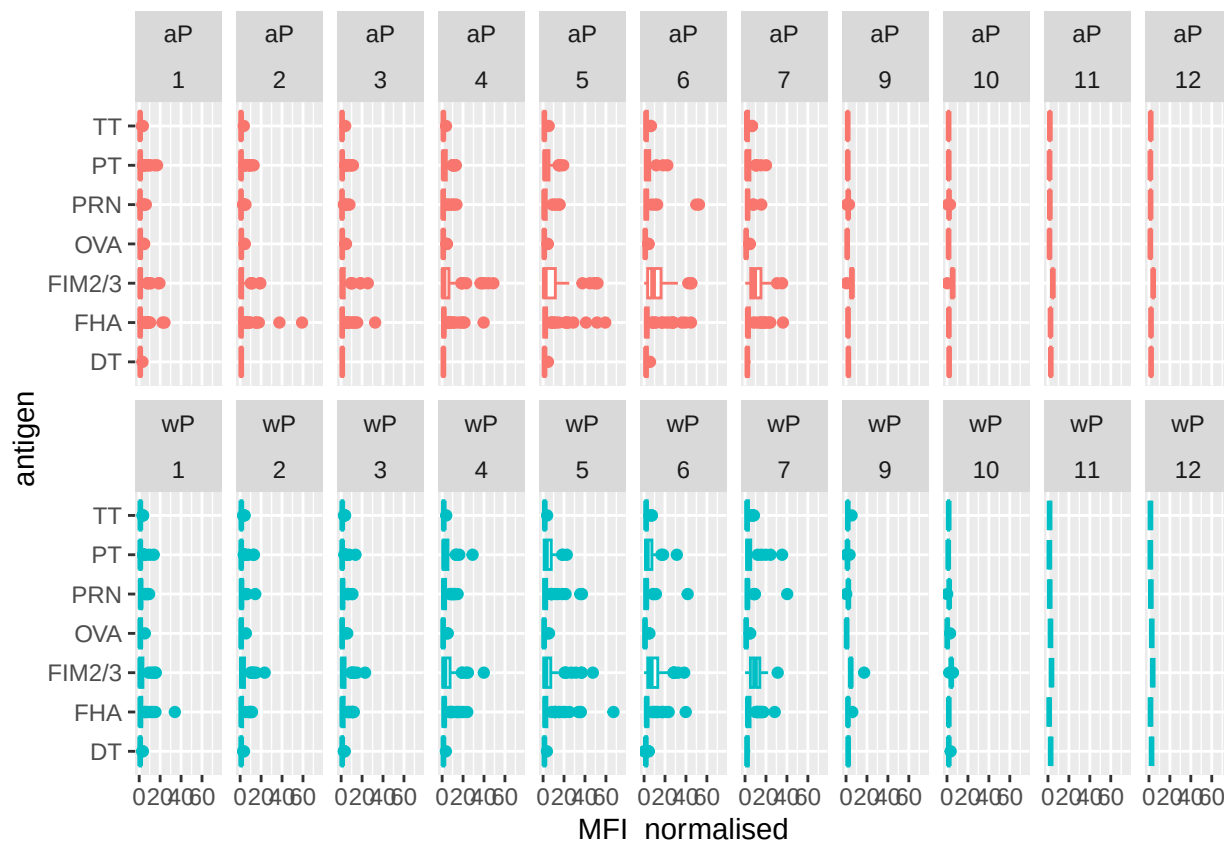
```
## Warning: Removed 5 rows containing non-finite outside the scale range
```

```
## (`stat_boxplot()`).
```



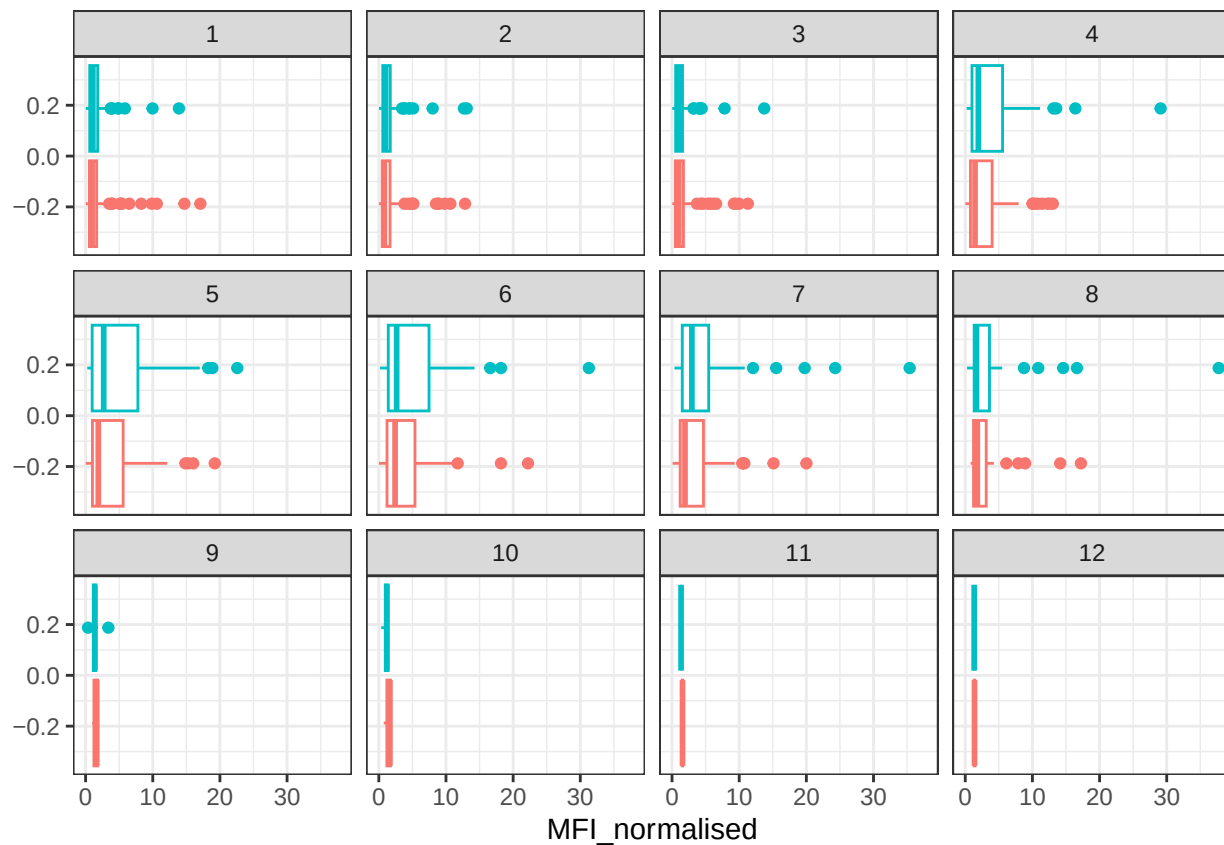
```
IgG %>% filter(visit != 8) %>%
ggplot() +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  xlim(0,75) +
  facet_wrap(vars(infancy_vac, visit), nrow=2)
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



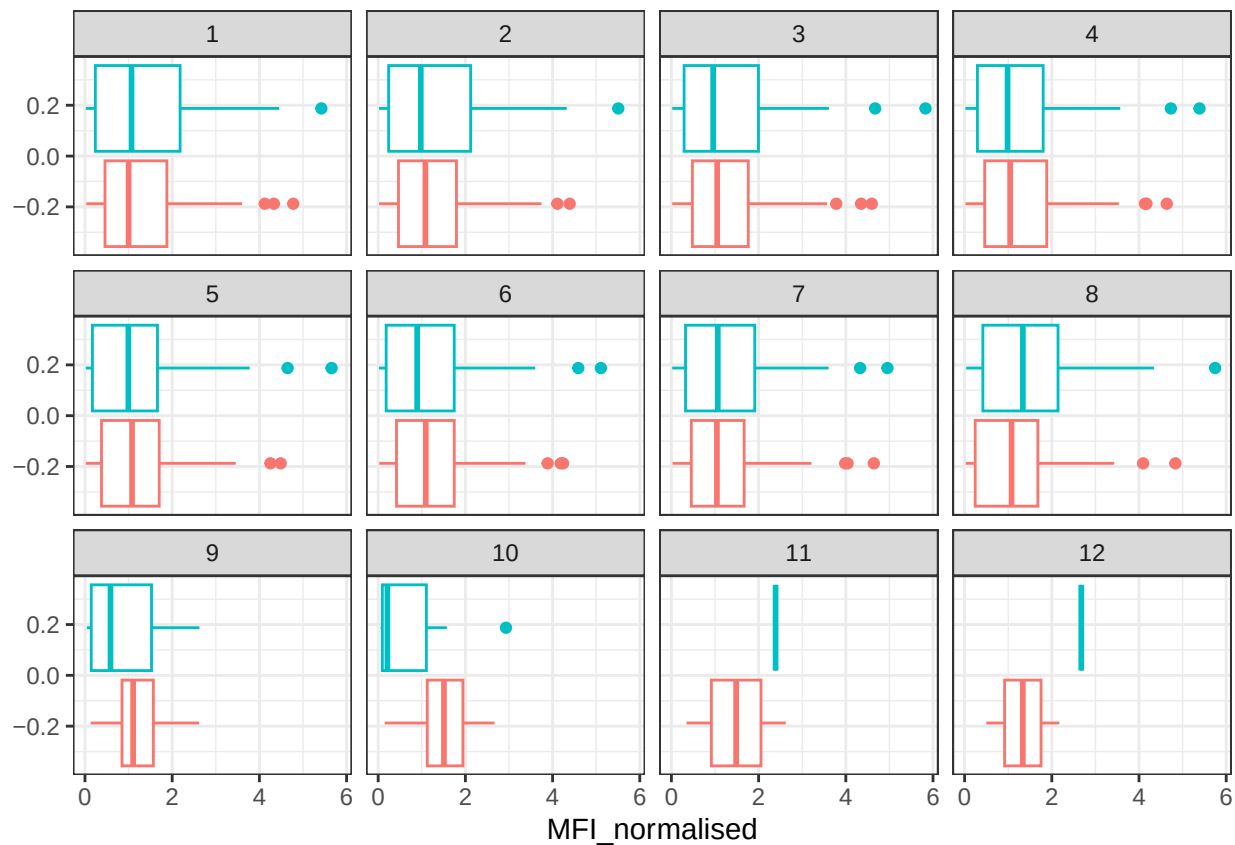
Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can choose any you like. Below I picked a “control” antigen (“OVA”, that is not in our vaccines) and a clear antigen of interest (“PT”, Pertussis Toxin, one of the key virulence factors produced by the bacterium *B. pertussis*).

```
filter(IgG, antigen=="PT") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



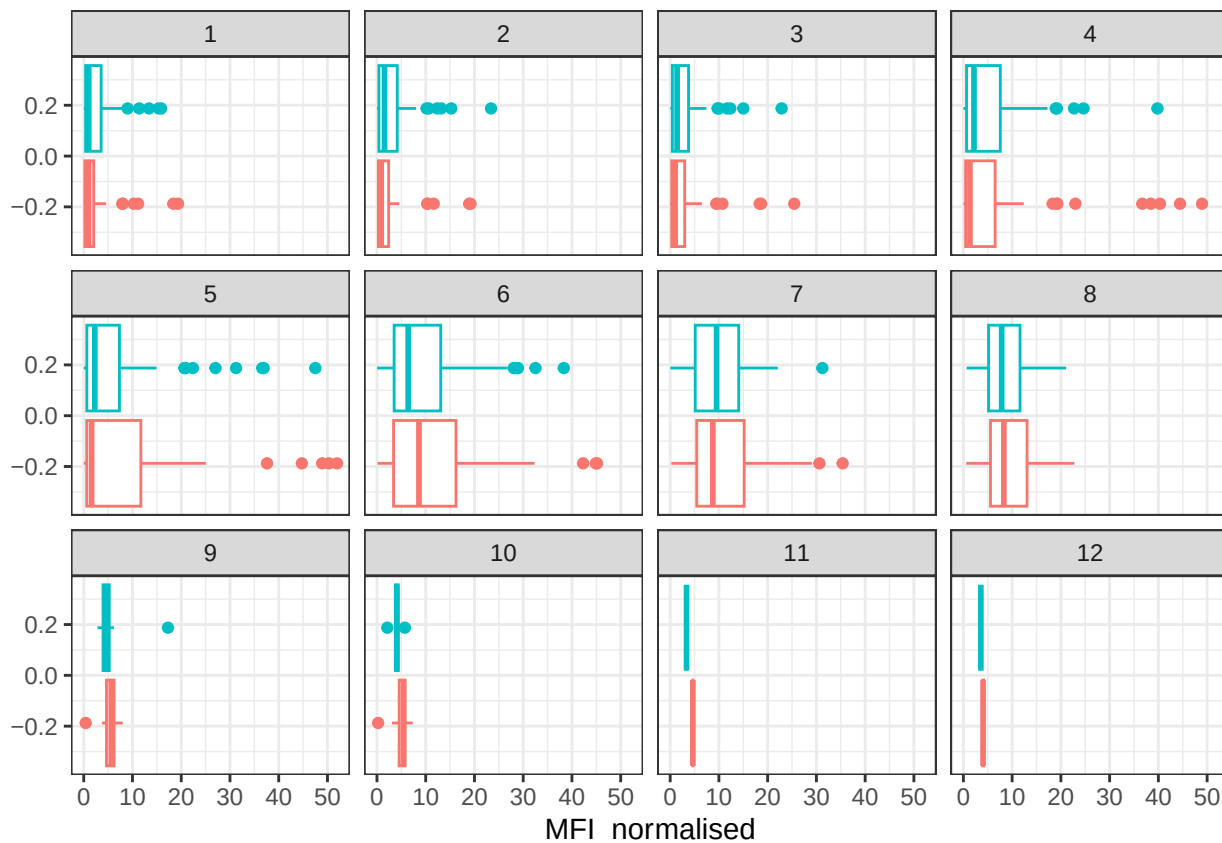
For OVA (The Control):

```
filter(IgG, antigen=="OVA") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



For FIM2/3 (The Vaccine Component):

```
filter(IgG, antigen=="FIM2/3") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



Q16. What do you notice about these two antigens time courses and the PT data in particular?

The PT antigen shows a strong increase after the booster shot, with the highest response around visit 5, which is about 14 days after the boost. After that, the antibody levels begin to decrease at later visits. In contrast, OVA stays low and mostly flat because it is a negative control and is not included in the vaccine.

Q17. Do you see any clear difference in aP vs. wP responses?

There is no very obvious difference between the aP and wP groups from this plot. Both groups seem to produce an immune response after vaccination, although there may be some variation between individuals.

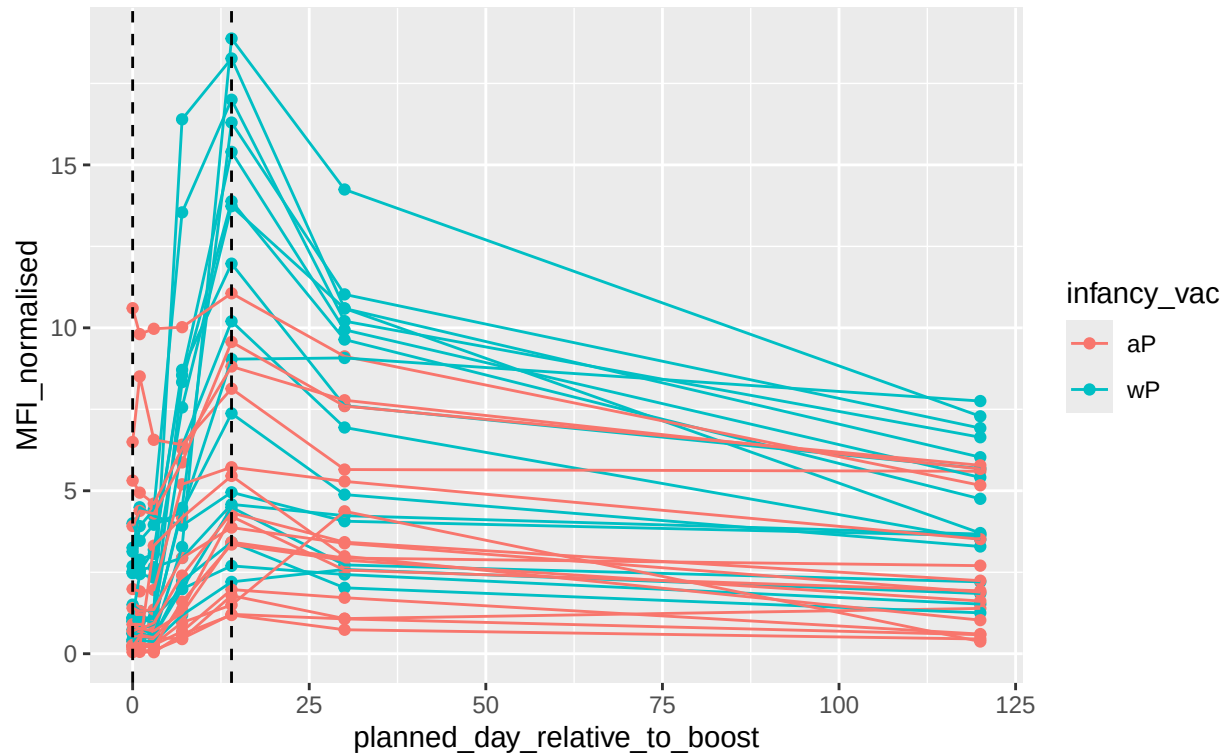
Q18. Does this trend look similar for the 2020 dataset?

```
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2021 dataset IgG PT",
         subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

2021 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)



The 2021 dataset shows a similar trend to the 2020 dataset, with IgG antibody levels against PT increasing after the booster and reaching a clear peak around day 14. After the peak, antibody levels gradually decline over time. The wP group generally appears to have higher PT antibody responses than the aP group, although both groups show the same overall pattern of increase and decline.